JENNIFER HSIA

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Machine Learning Ph.D. candidate at CMU working on LLM evaluation, model behavior, and reliability. Builds controlled evaluations that expose when models draw conclusions their inputs don't justify, and develops methods that reduce these failures.

EDUCATION

Carnegie Mellon University 2021–Present
Ph.D. in Machine Learning (expected May 2027) · M.S. in Computer Science (2023)
Advisors: Zachary C. Lipton and Aarti Singh

Princeton University 2017–2021
B.S.E. in Computer Science, magna cum laude · Minors in Statistics and Machine Learning

RESEARCH EXPERIENCE

Carnegie Mellon University | Graduate Researcher, Machine Learning Department 2021–Present

Deference to user claims in LLM advice In submission

- Studying how users' claims about their own circumstances shape LLM advice.

Robust answer selection when retrieval favors a wrong answer EMNLP 2026

- Showed that retrieved evidence can make a wrong answer look compelling even when it misses a key constraint in the question, and, in controlled experiments, that adding passages supporting such answers makes LLMs more likely to choose them.
- Developed and implemented Joint Constraint Aggregation (JCA), which extracts shared constraints from a question and jointly evaluates candidate answers against them, rather than scoring candidates independently. Built the evaluation pipeline over web-retrieved evidence.
- JCA was the most accurate method in all 9 combinations tested (3 instruction-tuned open-weight models × 3 evaluation settings), by 2–11 points over the best baseline, and lost less accuracy than any standard baseline when retrieval favored a wrong answer.

Robustness of RAG systems to the number of retrieved passages ICML 2025

- Showed that single-depth RAG evaluations hide brittleness: as more passages are added, some models peak early and degrade, while others improve and plateau.
- Introduced metrics for stability (holding near peak performance) and scalability (gains from more passages). Models more easily distracted by irrelevant passages tended to peak early and degrade, a pattern that held across retrievers, rerankers, and prompts.
- Led the experimental design and analysis, and implemented the retrieval and reranking pipelines (sparse, dense, and cross-encoder).

Gaming explanation-faithfulness benchmarks Findings of EACL 2024

- Showed that explanation-faithfulness benchmarks can be gamed to produce high scores without making explanations any more faithful.
- On ERASER, a simple wrapper that detects the benchmark's masked inputs inflated scores sharply without changing predictions or explanations on real inputs. On EVAL-X, encoding the predicted label in the explanation achieved the maximum score, though the benchmark was designed to block this shortcut.

INDUSTRY EXPERIENCE

Meta | Research Scientist Intern, Ads Ranking May–August 2025

- Developed a compact representation of long user-behavior histories for ads ranking: older history is compressed while recent events are kept in full, to fit memory and serving-throughput limits.
- Trained ranking models with this representation. In offline evaluation on recent data, it improved normalized entropy by 0.03% relative, above the internal threshold for a meaningful gain, at under 2% throughput cost.

PUBLICATIONS

** equal contribution*

When the Rule Is Clear but the Case Isn't: How User Claims Shape LLM Advice
Jennifer Hsia, Zachary C. Lipton, Aarti Singh. *In submission*

Joint Constraint Aggregation for Qualifier-Sensitive Retrieval-Augmented QA
Jennifer Hsia, Zachary C. Lipton, Aarti Singh. *EMNLP 2026* · Paper · Code

RAGGED: Towards Informed Design of Scalable and Stable RAG Systems
Jennifer Hsia*, Afreen Shaikh*, Zhiruo Wang, Graham Neubig. *ICML 2025* · *Earlier version: oral, NeurIPS 2024 Workshop on Adaptive Foundation Models* · Paper · Code

Goodhart's Law Applies to NLP's Explanation Benchmarks
Jennifer Hsia, Danish Pruthi, Aarti Singh, Zachary C. Lipton. *Findings of EACL 2024* · Paper · Code

An Experimental Study of Balance in Matrix Factorization
Jennifer Hsia, Peter Ramadge. *CISS 2021* · Paper

TECHNICAL SKILLS

Programming & ML: Python, PyTorch, Hugging Face Transformers, PEFT (LoRA), vLLM

Infrastructure: Slurm, GPU training and inference

TEACHING & SERVICE

Teaching Assistant, CMU: Human-AI Complementarity for Decision Making (2025), Probabilistic Graphical Models (2023)

Machine Learning Department TA Award, 2023 (one of four TAs recognized for 2021–2023)

Workshop Organizer: Human Evaluation of Generative Models, NeurIPS 2022

Reviewer: ACL, EMNLP, ICML